EXPERIMENT DESIGN AND MATERIALS · STAGES 0 AND 1

Two experiments, and what to order

The first two stages of the Hamiltonian programme, counted rather than estimated. Both come in well under budget, and three things about them changed once the theory pipeline was run.

What these two stages decide

Stage 0 measures how reproducible a single designed hinge is. Not its free energy — the *spread* across nominally identical constructs. That spread is the manufacturing noise floor, and it sets the resolution of everything after it.

Stage 1 is the decision point of the entire programme. It measures the coupling between two hinges and then asks the question that matters: **is that coupling constant when the biases are varied?** If it is not, there is no Hamiltonian, only a coupled mess, and the programme stops. A negative result here is a real finding about molecular coupling and is publishable.

Three changes from the plan

All three came out of running the analysis on synthetic data before ordering anything, which is what that pre-stage is for.

> 1 · STAGE 0'S GATE WAS THREE TIMES TOO LOOSE
>
> The plan asks that construct-to-construct spread be *below about 1 kT*. Simulated: at `1 kT` of spread, a designed `1 kT` coupling comes back with a median error of `0.85 kT`. Stage 1 then cannot meet its own requirement that J be constant to `0.3 kT`, whatever the sampling depth.
>
> **The gate should be 0.3 kT.** A 1 kT result would read as a pass and would have sent us into a stage that could not succeed.

2 · THE LIGATION CALIBRATION WAS AIMED AT THE WRONG THING

The plan calls for constructs locked in known states to measure ligation efficiency *per site*, because a state-dependent efficiency is warned to corrupt every fitted parameter.

It cannot corrupt J. A bias depending on each spin separately multiplies every configuration's weight by a per-spin factor, which is **algebraically identical to a shift in the bias fields** — and fields do not enter J at all. Simulated at biases up to 0.6 in log-odds, uniform across sites or different at each: the recovered coupling moves by `0.001 kT`.

What *does* reach J is an efficiency that depends on the **joint** state — different when the two hinges agree. That cannot be absorbed into fields, and it adds exactly **half itself** to the apparent coupling: a bias of 0.5 turns a true J of 1.0 into a measured 1.25.

So the locked controls should be locked in **all four joint states**, not in the two states of one spin. This is cheaper than the original control, not more expensive.

3 · SAMPLING DEPTH IS NOT THE CONSTRAINT

An undesigned three-body term of `0.4 kT` is detectable at `1000` molecules, `0.1 kT` at `10 000`, and `0.05 kT` at `100 000`. Sequencing delivers around 10^9.

Four orders of magnitude of margin. Effort spent on read depth is wasted; the systematics — ligation bias and PCR chimeras — are what will limit the result.

Stage 0 — the forward map, one spin

10 constructs, sharing a single staple core so that only the latch changes. Six latch lengths at balanced composition, then four GC fractions at a fixed `8 bp` length — the two axes swept separately, because swept together a change in free energy could be either and the stage exists to tell them apart.

Measure: bulk FRET giving p(up) against temperature (25–45 °C) and magnesium (8, 12, 16 mM). At least three independent folds of each construct, and at least two independent oligo syntheses for a subset — without which folding variance and synthesis variance cannot be separated, and separating them is the point. AFM on each to confirm it folded at all before its free energy is believed.

> GO / NO-GO
>
> Construct-to-construct spread at fixed design below **0.3 kT**. Above it, couplings of 1–3 kT are unresolvable, the mechanical coupling route is dead, and the choice is the stronger strand-mediated coupling or stopping.

Stage 1 — two spins

25 constructs. The full factorial is 54; the grid is complete only where it decides something. Six coupling mechanisms — three mechanical strut lengths, one steric, two strand-mediated tether lengths — with the full 3×3 bias grid on the medium strut, and corners elsewhere.

Measure: two-colour smFRET giving the full 2×2 joint distribution, from which J follows exactly as a log-odds ratio. HS-AFM on the best variant for flipping kinetics. oxDNA umbrella sampling on the same construct, as a third description that agrees or disagrees independently.

The key plot is fitted J against applied bias. A horizontal line is success; any slope means the couplings are bias-dependent and the pairwise form is wrong.

Controls, all three mandatory

- **Uncoupled pair at every bias setting** — two hinges far apart on the same plate, no strut. Any correlation there is an artefact, and it quantifies the background |J| must beat.
- **Constructs locked in all four joint states** — see change 2 above.
- **Two initial conditions for every measurement** — annealed from high temperature, and released from a locked state — confirming they converge. Otherwise the measurement is of kinetics, not of a Boltzmann distribution, and nothing in the numbers would say so.

> GO / NO-GO
>
> |J| above the Stage 0 noise floor by 3σ; J constant across the bias grid to `0.3 kT`; the uncoupled control below the noise floor. If J drifts with bias, stop and reconsider the coupling mechanism.

Oligos

The shared core is the entire economy of this. As independent constructs the same two series would cost `€36 800`.

STAGE	CONSTRUCTS	CORE STAPLES	VARIANT STAPLES	MODIFIED	COST
0 — one spin	10	200	120	2	€1324
1 — two spins	25	380	800	4	€4376
Both	**35**	**580**	**920**	**6**	**€5700**

THE FRET PAIR GOES ON THE CORE, AND IT MATTERS

Fluorophore-modified oligos cost roughly fifty times a plain one, and usually need HPLC purification. Put a donor/acceptor pair on each construct and Stage 0 gains €3000, tripling it.

But the pair sits *across the hinge joint*, which is core geometry rather than latch. So two modified oligos serve the whole series: €300 instead of €3000. Stage 1 needs two colour pairs, because one cannot tell which of the two hinges flipped.

Everything else

ITEM	FOR	NOTE
M13mp18 scaffold	every fold	one preparation covers both stages many times over
Tris, EDTA, $MgCl_2$	folding buffer	5 mM / 1 mM / screened 5–20 mM, pH 8
Agarose, TBE, stain	fold check	every construct, before anything else is believed
PEG 8000 or 100 kDa spin filters	purification	free staples on mica make counting unreliable
Thermocycler with slow ramp	annealing	2D folds at 1 °C/min; a hinge is 3D and needs 1 °C/h overnight
Fluorimeter or plate reader	Stage 0 readout	temperature control is the requirement, not sensitivity
smFRET / TIRF	Stage 1 readout	partner instrument; gives the joint distribution and the kinetics
AFM, and HS-AFM	structure and kinetics	partner instrument
Mica, tape, ultrapure water, N_2	AFM sample prep	cheap, and the step most likely to be wrong

> BUDGET REAL TIME FOR FRET CALIBRATION
>
> Donor-only and acceptor-only controls, direct-excitation correction, the γ factor. Sloppy FRET calibration is the most common source of a wrong free energy, and Stage 0's entire output is a free energy.

Every number here is generated by the toolchain rather than typed: construct counts, staple counts, costs, detection thresholds and the noise floor all come from the same source that will produce the sequences. The three changes from the plan came from running the analysis pipeline on synthetic data before ordering anything — which is what that pre-stage exists for, and it has now paid for itself twice over in a gate that would have passed wrongly and a control aimed at the wrong quantity.